SARS

Respiratory diseases

The body uses various strategies to fight disease but sometimes illnesses arise that leave it, and the scientific world, dealing with exceptions. *Pneumonia* is a respiratory condition that causes an inflammation of lung tissue. The tissue firms up and fluid builds in the lungs, particularly in the alveoli. This causes problems with oxygen intake and the reduced oxygen levels may need to be supplemented using medical techniques such as the oxygen mask, which provides pure oxygen for the patient to inhale. Certain strains of bacteria and viruses are associated with typical pneumonia. There are around 50 types of pneumonia.

1 Outline the differences between bacteria and viruses.

2 Pneumonia caused by bacteria (eg *Streptococcus pneumoniae*) can be treated by antibiotics such as penicillin. Pneumonia caused by a virus cannot be treated with antibiotics. Explain why this is so.

Influenza is a highly contagious disease of the respiratory system caused by various strains of viruses. The symptoms include a fever, tiredness and aches and pains. It can occur almost anywhere in the world. The incubation period for influenza is 1 to 3 days and the illness lasts about 10 days.

3 What treatment is usually advised for someone suffering from flu?

Regularly new strains of influenza appear, many of which have been traced back to originating in China. When the disease affects large parts of the community, an *epidemic* can arise. Should the disease spread to other countries, then a *pandemic* results (*pan* all). In 1918/19 an outbreak of influenza (the 'Spanish flu') claimed 40 million lives. Soldiers returning from World War I in Europe are thought to have brought the disease to Australia, resulting in 12 000 deaths. Other large-scale infections have included the 'Asian flu' (1957) and the 'Hong Kong flu' (1968). Between 1995 and 2002, there were several outbreaks of influenza in Hong Kong and millions of chickens were inoculated or slaughtered as a result.

4 Give reasons why the authorities got rid of so many chickens at that time.

5 Outline the difference between an epidemic and a pandemic.

Atypical pneumonia: Severe Acute Respiratory Syndrome (SARS)

Severe, when referring to a disease, means that the symptoms are extreme or dangerous. An *acute* disease is one in which symptoms arise quickly and don't last long. When the signs are present to indicate that a patient has pneumonia, but the illness is not following the typical pattern of development, it is called *atypical pneumonia* (*atypical* unusual). In March 2003 the media informed the public about the appearance of an atypical pneumonia called *severe acute respiratory syndrome,* or SARS. The SARS outbreak was traced back to Guandong Province in China. It first appeared there in November 2002 but the authorities did not make it known and, within months, people visiting China had exported it to over a dozen countries. By late April 2003 SARS had reached 27 countries leaving 5000 affected and about 300 dead.

SARS was first believed to be a variant of the influenza virus but its incubation period is slightly longer – about 2 to 7 days (occasionally up to 10 days). While influenza has not been completely ruled out, other paths are being investigated, as researchers discovered a previously unrecognised *coronavirus* in patients with SARS.

6 Give a reason why authorities might be reluctant to report cases of a potentially fatal disease.

7 Give your own definition for *severe.*

8 Give your own definition for *acute.*

9 What does *'SARS was believed to be a variant strain'* mean?

The coronavirus gets its name from having a halo, or crown, around the virus particle (*corona* crown). The crown is formed by projections coming from the virion's outer coat and this is a distinctly recognisable feature of the *Coronaviridae* family. Coronaviridae contain a single strand of RNA and have different forms during the various stages of their lifecycle – this is called being *pleomorphic.* They are linked to respiratory illness, hepatitis (inflamed liver) and gastrointestinal infection. The condition caused by the virus, and its severity, depends on the animal species affected.

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Although previously unknown coronavirus has been isolated in SARS patients, however this does not mean that it is the main cause of the illness. Another RNA virus, the *paramyxovirus*, is also being investigated for possible links to SARS. Paramyxoviruses have a helical shape and the virus affects vertebrates.

Symptoms of SARS

Many of the symptoms associated with SARS might first be mistaken for flu. It begins with a fever of 38°C (104°F) or greater. Some people might also develop chills. Within 48 hours of the onset of fever, a dry cough develops. This leads to breathing difficulties and should not be ignored. Headache, muscle stiffness, loss of appetite and other flu-like symptoms can also develop but the fever and dry cough are indications that it could be more than flu. If the patient has also recently been overseas (particularly to Asia) or recently associated with someone who has, this information must be given to the doctor involved.

10 Name another illness with symptoms that might be mistaken for flu.

11 Suggest why so many illnesses share similar symptoms.

12 Viruses are often referred to as 'non-cellular'. Explain why.

Since viruses do not respond to antibiotics, there is no single cure for SARS. Doctors must treat the symptoms instead. These include giving antibiotics for the types of bacteria known to accompany typical pneumonia, as well as antiviral agents and whatever else the doctor feels will help to relieve the symptoms. At present, *droplet infection* (particles sneezed or coughed out) and contaminated objects (such as utensils) are considered the most likely cause of its spread.

13 Why do viruses not respond to antibiotics but bacterial agents do?

14 Since viruses do not respond to antibiotics, why would a doctor prescribe them to a SARS patient?

Many people get a vaccination for influenza every year just before the flu season begins. They can be those who have a greater risk of contracting influenza or who might suffer complications if they catch the disease. The vaccine used has to change at regular intervals because the strains of viruses that cause influenza can change rapidly. Common advice is to stay at home, rest, drink plenty of fluids and take aspirin or paracetamol.

15 Identify two groups of people who stand a greater than average change of catching influenza.

16 Identify two groups of people for whom catching influenza might pose a great health risk.

17 Give two reasons why a person with influenza might be advised to stay at home.

18 A person who had an effective vaccination against influenza might be totally unprotected a year later. Explain why this is.

Research

19 Explain the difference between a *virus* and a *viroid*.

20 Since this article was written, scientists will have learned much more about SARS. Outline any facts about SARS that are not mentioned here. (Hint: for more information about this, or any other illnesses that are globally spread, the *World Health Organization (WHO)* is a good place to start your search.)

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